INCH-POUND

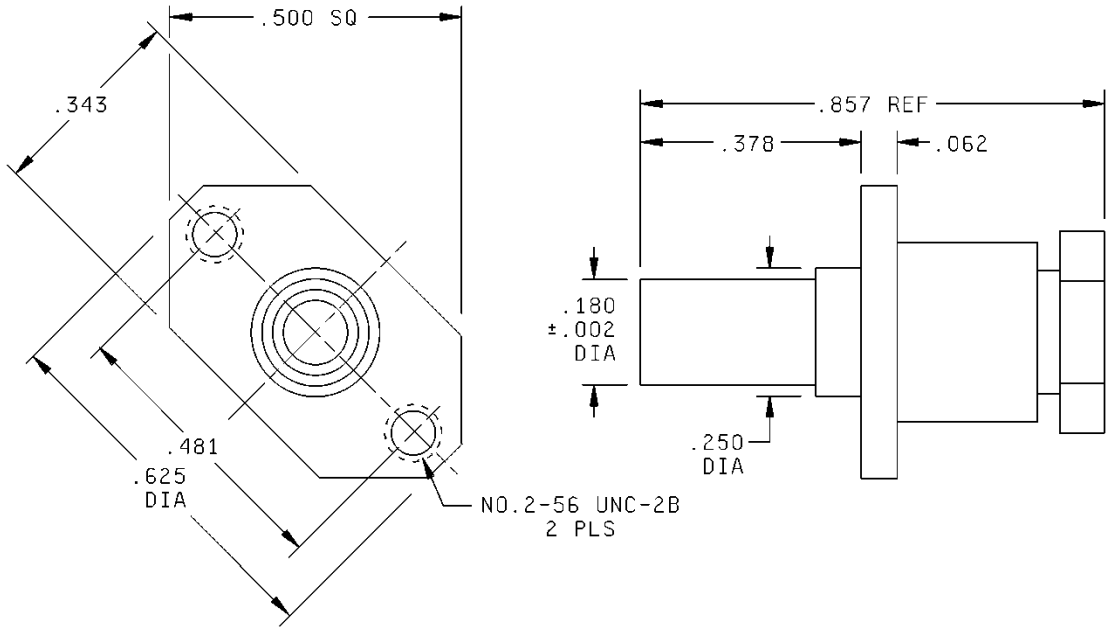
MIL-PRF-31031/40
w/AMENDMENT 2
16 February 2024
SUPERSEDING
MIL-PRF-31031/40
w/AMENDMENT 1
02 November 2018

PERFORMANCE SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, RECEPTACLE, RADIO FREQUENCY, HIGH RELIABILITY, COAXIAL, SERIES BMZ, SOCKET CONTACT, FOR FLEXIBLE CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-31031.



NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Unless otherwise specified all tolerances shall be ±.005 inch.
4. Interface series BMZ, socket contact, in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A FSC 5935

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TABLE I. Part or Identifying Number (PIN) to cable cross-reference.

PIN M31031/40	Cable accommodation M17/
A3N01	113-RG316 119-RG174 94-RG179 172-00001 173-00001
A3N02	60-RG142 84-RG223 128-RG400 158-00001 167-00001 175-00001
A3N03	152-00001

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: DC to 18 GHz.

Voltage rating: See table II.

Temperature rating: -65ºC to +165ºC.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Interface dimensions: In accordance with MIL-STD-348.

Force to engage:

Longitudinal force: 12 ounces.

Torque: Not applicable.

Coupling proof torque: Not applicable.

Mating characteristics:

Center contact (socket): Using hardened steel gage pins with a 4-8 micro surface finish and an insertion depth of .070 min the following forces shall be maintained:

Withdrawal force: 2 oz. min. with a .0195 +.0001/-.0000 dia. pin.

Insertion force: 12 oz. max, with a .0210 +.0000/-.0001 dia. pin.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

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Center contact retention: 6 pounds minimum axial force.

Radial torque: Not applicable.

Voltage standing wave ratio (VSWR): From .5 to 18 GHz. Or approximately 80 percent of the upper cutoff frequency of the cable, whichever is lower, see table III.

Moisture resistance: MIL-STD-202-106. No measurements shall be taken at high humidity. Insulation resistance shall be 200 megohms within 5 minutes after removal from humidity.

Contact resistance (in milliohms, maximum):

	Initial	After environmental
Center contact:	3.0	4.0
Outer contact:	2.0	N/A
Outer cable conductor to body:	0.5	N/A

Dielectric withstanding voltage: See table III.

Vibration, high frequency: MIL-STD-202-204 test condition E, except the method of mounting shall be approved by the qualifying activity. No discontinuity shall be permitted.

Corona level: See table III.

Insulation resistance: MIL-STD-202-302, 5,000 megohms minimum.

Shock (specified pulse, socket contact only): MIL-STD-202-213 test condition I.

Thermal shock: MIL-STD-202-107 test condition B (except high temperature to be +115°C).

Corrosion (salt spray): MIL-STD-202-101 test condition B.

RF high potential withstanding voltage: See table III.

Frequency: 5 to 7 MHz.

Leakage current: Not applicable.

RF leakage: See table III.

RF insertion loss: See table III.

Cable retention: Shall be specified in table IV.
Coupling mechanism retention force: Not applicable.

Durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

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TABLE II. Connector voltage rating.

PIN M31031/40	Voltage max. at sea level (V rms)	Voltage max. at 70,000 ft. (V rms)
A3N01	250	65
A3N02	335	85
A3N03	250	60

TABLE III. Electrical performance characteristics.

Parameter	PIN M31031/40		
	A3N01	A3N02	A3N03
VSWR	1.15 +.01F (F in GHz)	1.15 +.01F (F in GHz)	1.15 +.02F (F in GHz)
Insertion loss in dB	.06 √F (F in GHz)	.06 √F (F in GHz)	.06 √F (F in GHz)
DWV V rms At sea level	750	1000	750
Corona V rms at 70,000 ft.	190	250	190
RF high pot. V rms	500	670	500
RF leakage in dB	-60	-60	-60

TABLE IV. Cable retention forces.

PIN M31031/40	Axial force (minimum, lbs.)		Torque (minimum, inch-ounces)
	Flexible		
	Single braid	Double braid	
A3N01	20	N/A	N/A
A3N02	40	45	N/A
A3N03	25	25	N/A

PIN: See table I.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-31031, this document references the following:

MIL-STD-202-101
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-202-302
MIL-STD-348

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2024-045)

Review activities:
Army - AT, AV, EA, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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